

Test V1: (1h:30)

Full Name:

Note: Show all the solution steps

Group:

Exercise 1: (1 pts)

Question	Answer
How many values can a 6-bit unsigned number represent?	
Calculate the maximum value that can be represented by a 10-bit unsigned number?	
How many bits are required to represent the decimal number 2047 in binary?	
How many bits are needed to represent the number -128 in Two's Complement ?	

Exercise 2: (6pts)

1. The following conversions contain various types of **mistakes**. Identify the **errors** in each conversion. Provide the correct conversions. show all steps calculation. Take the 1st one as reference and check the others:

- $(25)_{10} = (11000)_2 = (41)_8 = (19)_{16}$
- $(145.625)_{10} = (10011001.100)_2 = (221.5)_8 = (99.4)_{16}$
- $(111001.011)_2 = (57.5)_{10} = (11.3)_8 = (39.6)_{16}$
- $(141.3030)_5 = (101110.100)_2 = (46.625)_{10} = (55.44)_8$
- $(123.3)_4 = (11010.11)_2 = (27.75)_{10} = (55.B)_{16}$
- $(91.5)_{10} = (5B.8)_{16} = (321.3)_6 = (133.4)_8$

Check and correct:

2. Perform the following operations in binary and verify the result using Decimal equivalents.

$(101011.101)_2 + (1110.1011)_2$	$(3E.8)_{16} \times (7.3)_8$
$(1111101.110)_2 - (4F.2)_{16}$	

Exercise 3: (1pts)

Determine the values of b_1 and b_2 , which are the bases of the numbers $(233)_{b_1} = (442)_{b_2}$ respectively. Choose the correct option and justify your answer:

a. $b_1 = 7$ and $b_2 = 5$.

b. $b_1 = 8$ and $b_2 = 6$.

c. $b_1 = 6$ and $b_2 = 4$.

